Open Source Architecture Appendix

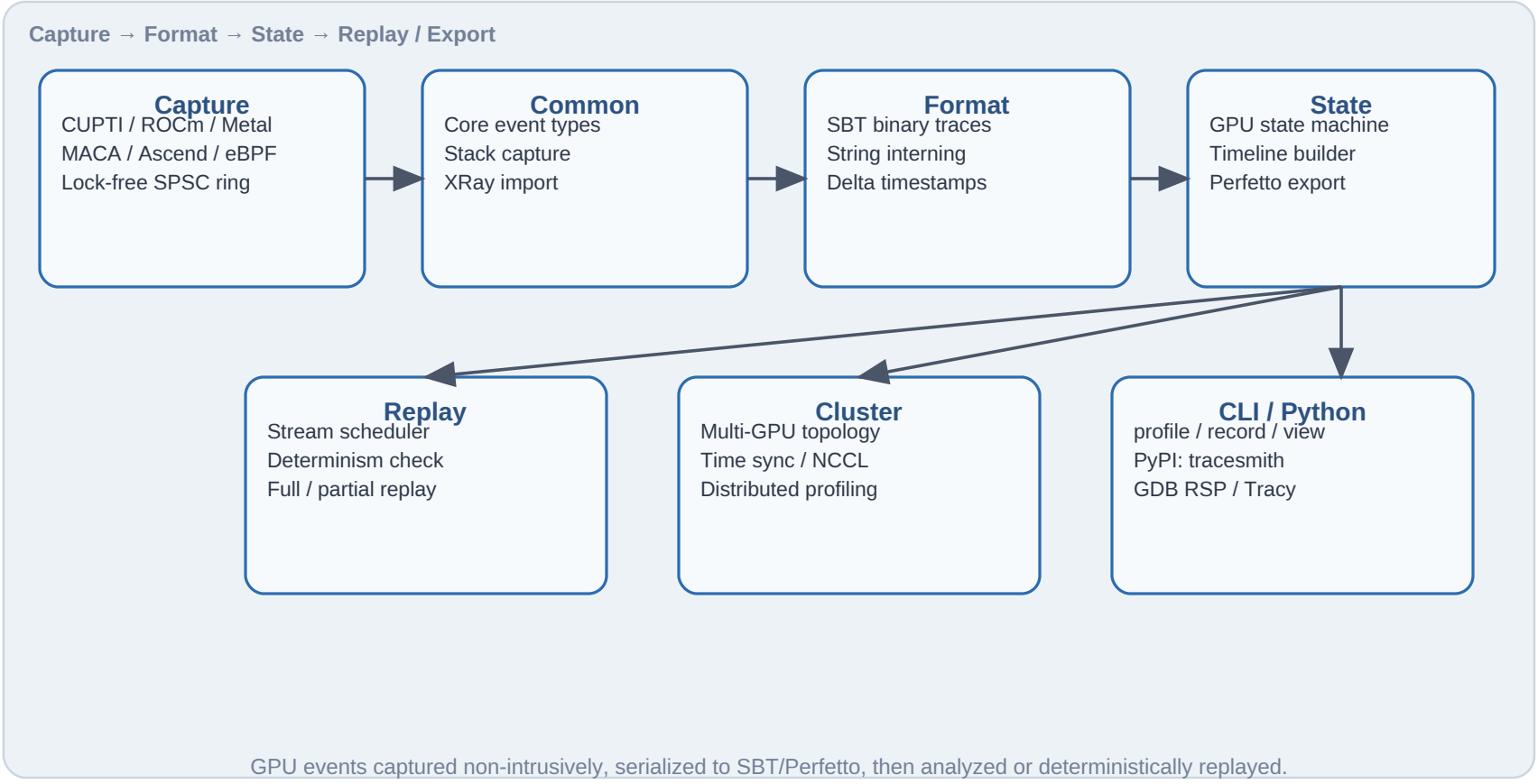
Chen Xingqiang · github.com/chenxingqiang

TraceSmith · YiRage · JAX-sklearn · Secret-Learn

YouTube-DNN-RecSys · Research CLI · Questions-Gen

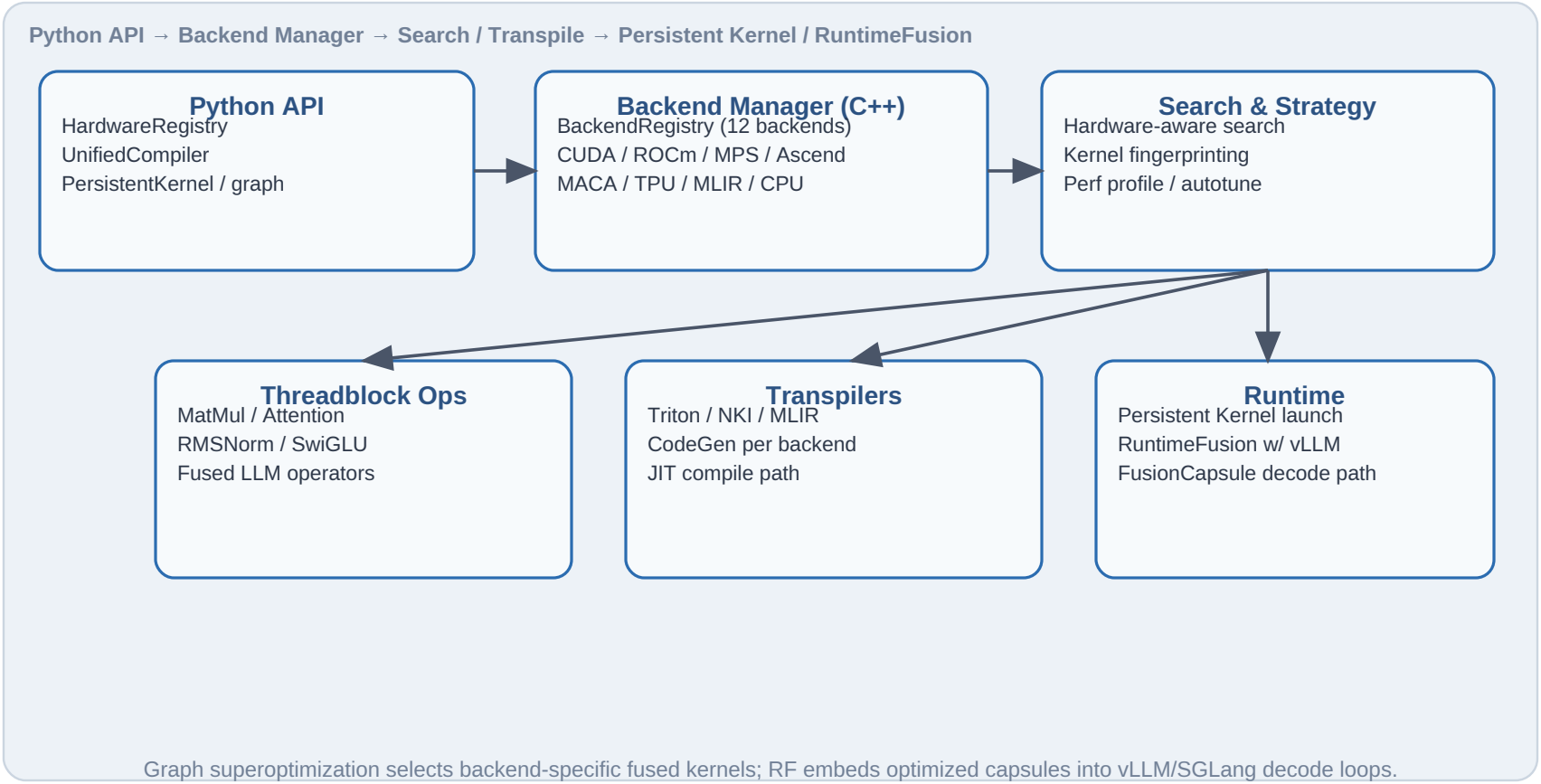
TraceSmith

Cross-platform GPU Profiling & Replay · github.com/chenxingqiang/TraceSmith



YiRage (YPK)

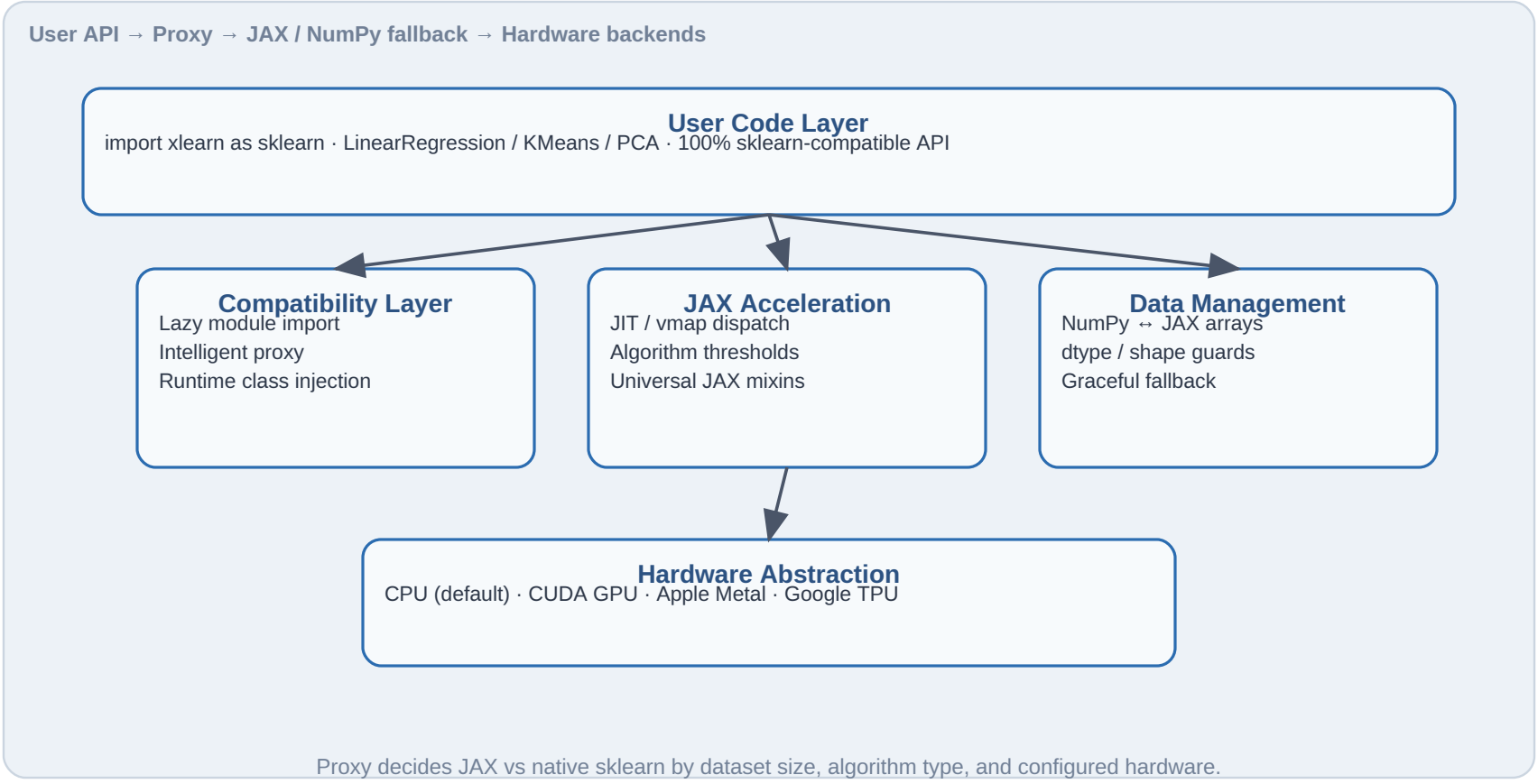
Multi-backend LLM megakernel compiler & runtime · github.com/chenxingqiang/YiRage



Graph superoptimization selects backend-specific fused kernels; RF embeds optimized capsules into vLLM/SGLang decode loops.

JAX-sklearn (xlearn)

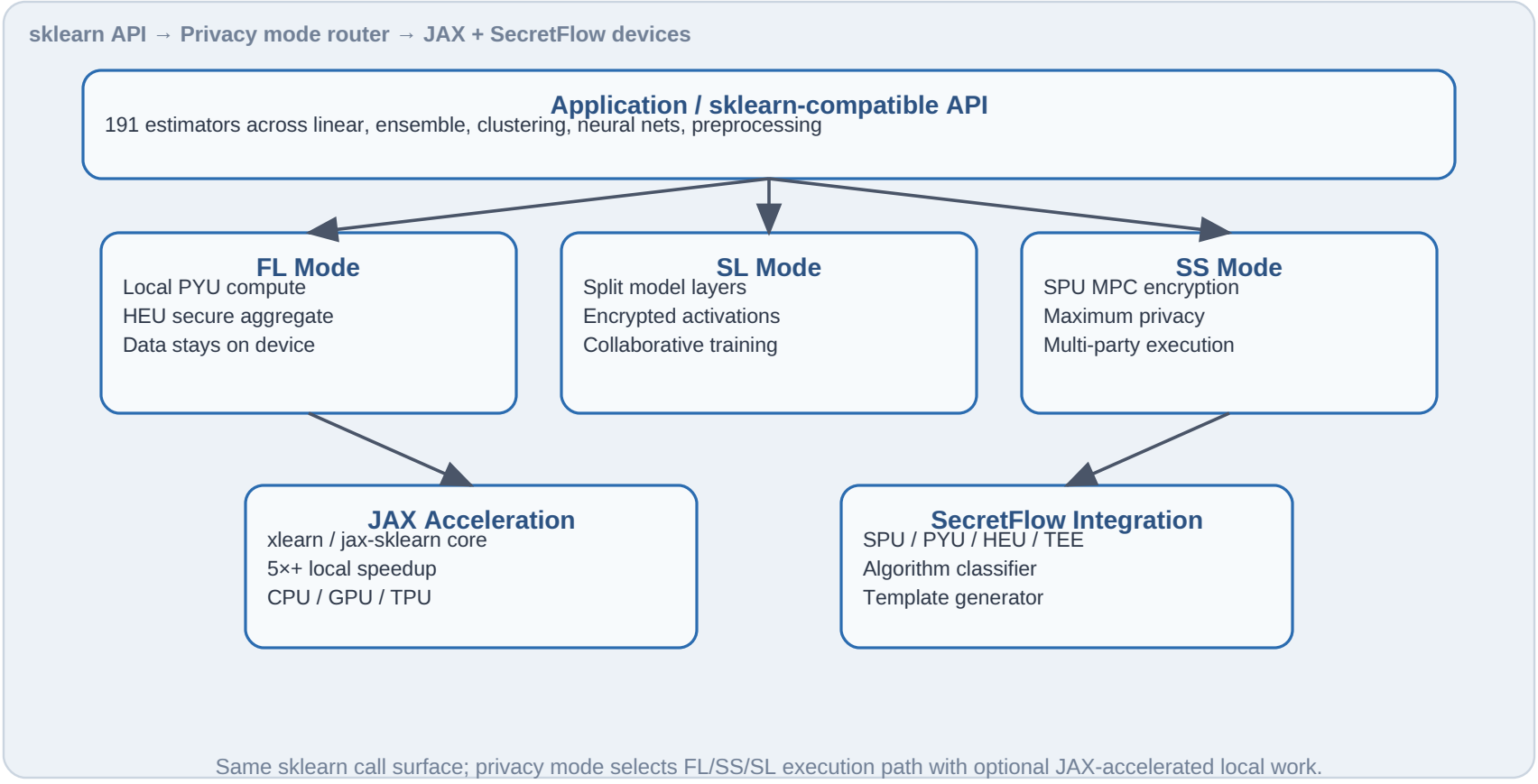
Drop-in sklearn API with automatic JAX acceleration · github.com/chenxingqiang/jax-sklearn



Proxy decides JAX vs native sklearn by dataset size, algorithm type, and configured hardware.

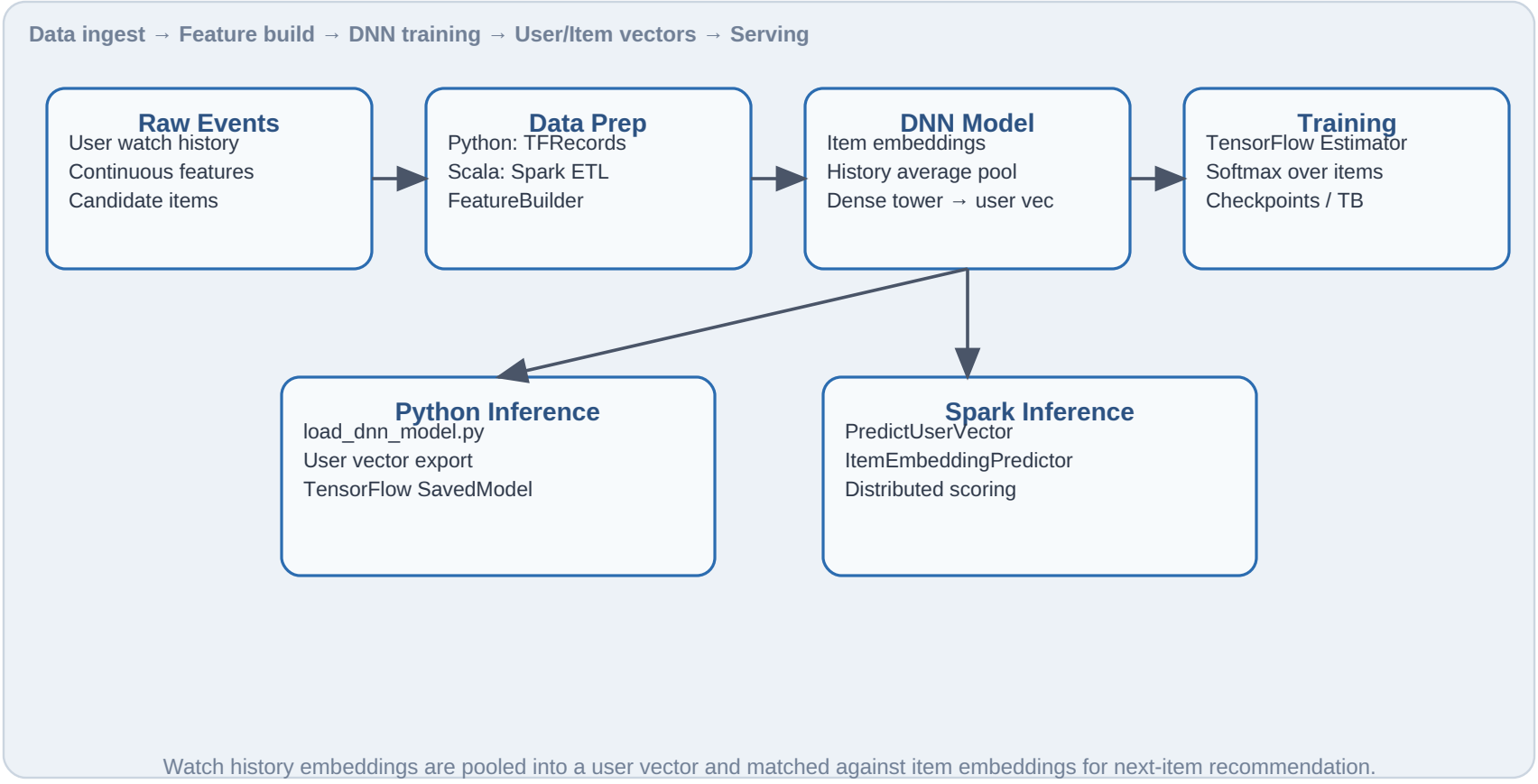
Secret-Learn

Privacy-preserving ML · 191 algorithms × FL/SS/SL · github.com/chenxingqiang/Secret-Learn



YouTube-DNN-RecSys

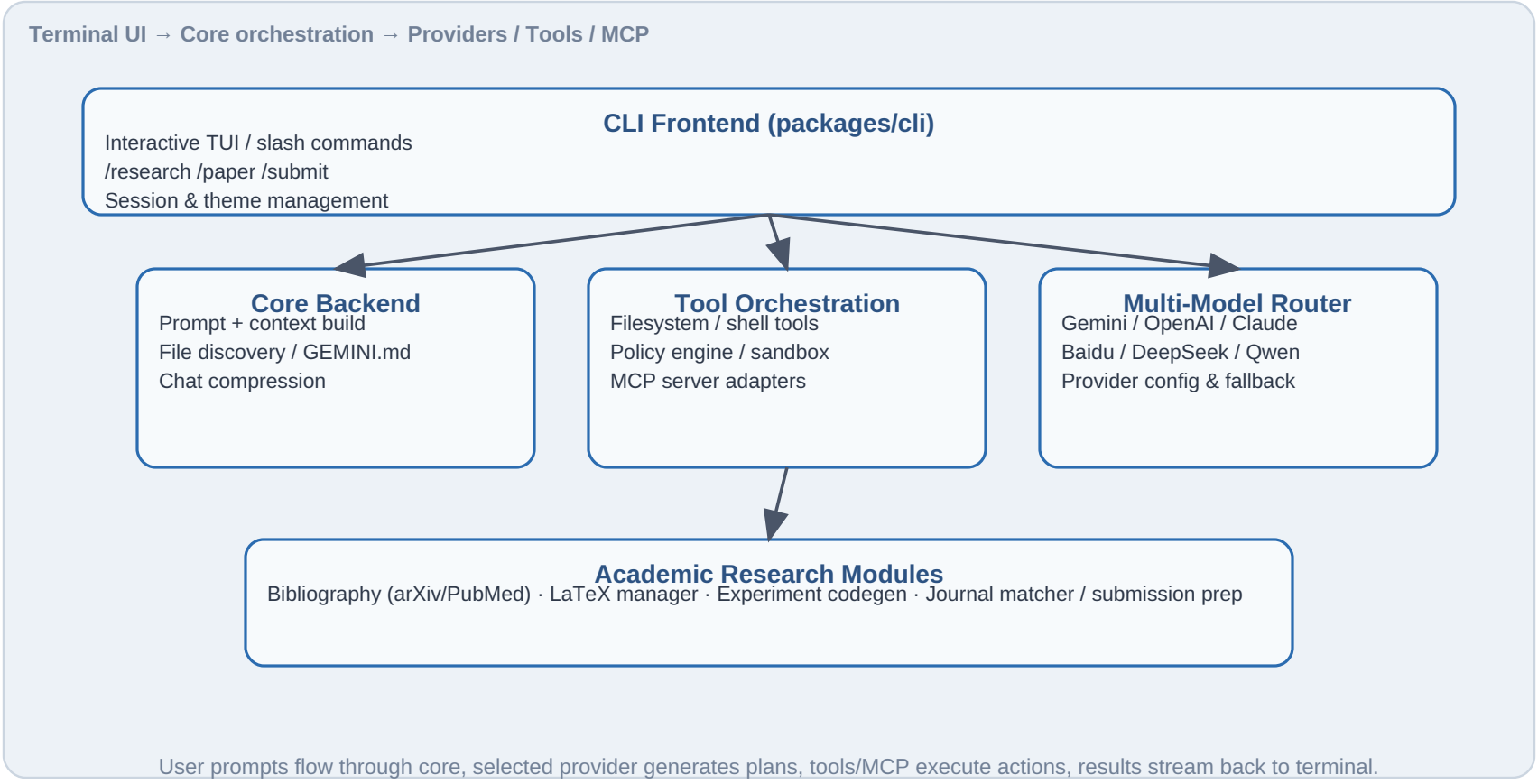
Two-tower recall pipeline · TensorFlow + Spark · github.com/chenxingqiang/YouTube-DNN-RecSys



Watch history embeddings are pooled into a user vector and matched against item embeddings for next-item recommendation.

Research CLI

CLI AI workflow + academic research tools · github.com/iechor-research/research-cli



Questions-Gen

Math competition problem generator · Qwen3-14B · github.com/chenxingqiang/GenQuestion

